Vanilla Options in Practice

Payoffs, Black–Scholes benchmarks, and static-arbitrage diagnostics

Lecture 9

One question drives the whole lecture

The question

Lecture 8 gave model-free principles for comparing claims. How do those principles discipline the prices of the options traders quote every day?

1. Build calls, puts, and simple option positions from their payoffs.
2. Recover put–call parity from replication, then use it to diagnose a mispriced quote.
3. Read Black–Scholes as a pricing benchmark and a language for risk.
4. Turn dominance, monotonicity, and convexity into quote checks.
5. Construct the trade whenever a check fails.

Payoffs and Market Language

A vanilla option is a simple right exercised at one date

At maturity T, the underlying trades at S_T and the strike is K.

European call

The right to *buy* one share for K:

$$\boxed{(S_T - K)^+.}$$

It is exercised only when $S_T > K$.

European put

The right to *sell* one share for K:

$$\boxed{(K - S_T)^+.}$$

It is exercised only when $S_T < K$.

NO-ARBITRAGE "European" fixes the exercise date; it does not describe where the option trades or who may hold it.

Can you fill in the two payoff tables?

YOUR TURN Take $K = 100$. Fill in the call and put payoffs when $S_T = 80$, 100, and 130. Which contract protects a fall in the stock price?

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S_T	80	100	130
Call $(S_T - 100)^+$	0	0	30
Put $(100 - S_T)^+$	20	0	0

Reading the table

The call has upside exposure; the put is downside insurance.

Going short flips every payoff

Long position

Buying the call gives $(S_T - K)^+$. Buying the put gives $(K - S_T)^+$.

Short position

Selling the same contracts gives $-(S_T - K)^+$ and $-(K - S_T)^+$. The premium is received today; the obligation is carried to T.

NO-ARBITRAGE For no-arbitrage arguments, write the *net* payoff of the whole trade, including every long and short leg.

Moneyness says whether immediate exercise has value

At time t, compare spot S_t with strike K.

	In the money	At the money	Out of the money
Call	$S_t > K$	$S_t \approx K$	$S_t < K$
Put	$S_t < K$	$S_t \approx K$	$S_t > K$

Important distinction

Moneyness uses S_t, but the European option payoff is determined by S_T. An OTM option today may finish ITM.

Which option has the most time value?

YOUR TURN Two one-year calls have the same spot $S_t = 100$, volatility, and rate. Their strikes are 80, 100, and 130. Which one is most likely to consist mainly of time value rather than intrinsic value?

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Answer

The $K = 100$ call is at the money. Its intrinsic value is zero, so nearly all of its price is time value: the value of possible future moves. Deep ITM and deep OTM options typically have less time value.

A straddle buys movement in either direction

A long straddle is one call plus one put, both with strike K:

$$(S_T - K)^+ + (K - S_T)^+ = \boxed{|S_T - K|}.$$

When S_T is near K

Both options expire with little or no payoff.

When S_T moves far

One option pays, and the payoff grows with the distance from K.

NO-ARBITRAGE The payoff is convex: uncertainty has value to the holder of the straddle.

Stock plus put creates a floor; stock minus call creates a cap

Protective put

$$S_T + (K - S_T)^+ = \max(S_T, K).$$

The investor keeps upside but cannot finish below K.

Covered call

$$S_T - (S_T - K)^+ = \min(S_T, K).$$

The investor gives up upside above K for the call premium.

Put–Call Parity

Two portfolios produce the same terminal wealth

Assume no dividends and let $D_t = e^{-r(T-t)}$. Compare:

Portfolio A

One call plus cash growing to K:

$$(S_T - K)^+ + K.$$

Portfolio B

One put plus one share:

$$(K - S_T)^+ + S_T.$$

YOUR TURN Check both portfolios for $S_T \leq K$ and for $S_T > K$. What common payoff do they deliver?

The common payoff is $\max(S_T, K)$

	$S_T \leq K$	$S_T > K$
Call + cash	$0 + K = K$	$(S_T - K) + K = S_T$
Put + stock	$(K - S_T) + S_T = K$	$0 + S_T = S_T$

NO-ARBITRAGE The law of one price gives

$$\boxed{C_t + KD_t = P_t + S_t, \qquad C_t - P_t = S_t - KD_t.}$$

Put–call parity is a model-free identity

$$\boxed{C_t - P_t = S_t - Ke^{-r(T-t)}.}$$

What parity does not need

No lognormal distribution, volatility estimate, or Black–Scholes formula. Only the same underlying, the same strike, the same maturity, and a tradable funding account.

WATCH OUT With continuous dividend yield q, replace S_t by $S_t e^{-q(T-t)}$. The no-dividend formula then creates a false arbitrage signal.

Can parity recover a put quote?

YOUR TURN Let $S_t = 100$, $K = 100$, $T - t = 1$, $r = 5\%$, and $C_t = 10.45$. Use put–call parity to compute the put price.

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$$Ke^{-r(T-t)} = 100e^{-0.05} \approx 95.12,$$

$$P_t = C_t - S_t + Ke^{-r(T-t)} = 10.45 - 100 + 95.12 \boxed{\approx 5.57.}$$

Reading the result

Once three of the four inputs are known, parity fixes the fourth.

A parity violation tells us exactly what to trade

Suppose the market quotes $C_t = 12$, $P_t = 5$, $S_t = 100$, and $Ke^{-r\tau} = 95.12$.

$$\underbrace{C_t + Ke^{-r\tau}}_{107.12} > \underbrace{P_t + S_t}_{105}.$$

YOUR TURN Which portfolio is expensive? Specify a trade that produces cash today and zero net payoff at maturity.

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YOUR TURN Which portfolio is expensive? Specify a trade that produces cash today and zero net payoff at maturity.

Trade

Sell the call and borrow 95.12; buy the put and the share. The initial receipt is $107.12 - 105 = 2.12$. The two sides have identical terminal payoff.

Black–Scholes as a Benchmark

Black–Scholes adds a model to the no-arbitrage framework

MODEL Under the pricing measure $\mathbb{Q}$, assume constant r, constant volatility σ, no dividends, and

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}.$$

What the model contributes

It specifies a distribution for S_T, so it produces a numerical price. No-arbitrage alone gives relations and bounds, but not generally a unique price for every claim.

The call formula separates stock value from strike cost

Let $\tau = T - t$. The Black–Scholes call is

$$\boxed{C_t = S_t \Phi(d_1) - K e^{-r\tau} \Phi(d_2),}$$

where

$$d_1 = \frac{\log(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

A useful interpretation

$\Phi(d_2) = \mathbb{Q}(S_T > K \mid \mathcal{F}_t)$ in this model: the risk-neutral chance that the call finishes in the money.

Can you price an at-the-money one-year call?

YOUR TURN Use $S_t = K = 100$, $r = 5\%$, $\sigma = 20\%$, and $\tau = 1$. Compute d_1, d_2, then use $\Phi(0.35) \approx 0.6368$ and $\Phi(0.15) \approx 0.5596$.

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$$d_1 = 0.35, \qquad d_2 = 0.15,$$

$$C_t = 100(0.6368) - 100e^{-0.05}(0.5596) \boxed{\approx 10.45.}$$

NO-ARBITRAGE The put price is then 5.57, either from its formula or from parity.

The Greeks turn a price into a local risk report

Greek	What it measures
Delta $\Delta = \partial_S V$	sensitivity to a small stock move
Gamma $\Gamma = \partial_{SS} V$	how rapidly Delta changes
Vega $\partial_\sigma V$	sensitivity to implied volatility
Theta $\partial_t V$	sensitivity to passage of time
Rho $\partial_r V$	sensitivity to the interest rate

For a Black–Scholes call

$$\Delta_{\text{call}} = \Phi(d_1), \qquad \Gamma = \frac{\Phi'(d_1)}{S_t \sigma \sqrt{\tau}}.$$

Delta is the first-order hedge ratio; Gamma is why that hedge must be rebalanced.

What does a call Delta of 0.64 mean?

YOUR TURN An option has $\Delta = 0.64$. For a small \$1 rise in the stock, what is the first-order change in the option price? How would you hedge a short position in 100 such calls?

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First-order answer

The option gains about \$0.64 per contract. A short position in 100 calls has Delta -64, so buying about 64 shares makes the portfolio locally Delta-neutral.

WATCH OUT Delta-neutral is not risk-free: Gamma, volatility, and discrete rehedging still matter.

Higher volatility raises the value of a long call and put

Why

The payoffs $(S_T - K)^+$ and $(K - S_T)^+$ are convex in S_T. Holding the mean fixed, a wider distribution puts more probability in the tails, where the holder benefits but cannot lose below zero at maturity.

NO-ARBITRAGE For standard European options, Vega is positive. A higher implied volatility therefore means a higher price, all else equal.

Implied volatility is the market price written in model units

Given a European call quote $C_t^{\mathrm{mkt}}(K, T)$, hold $S_t, K, \tau = T - t, r$ fixed and solve for the volatility:

$$C^{\mathrm{BS}}(S_t, K, \tau, r, \sigma_{\mathrm{imp},t}(K, T)) = C_t^{\mathrm{mkt}}(K, T).$$

Black–Scholes Vega is positive: the price increases with σ, so there is at most one solution. In practice, find it numerically.

Why traders quote volatility

It makes options with different strikes and maturities easier to compare than their dollar premiums alone.

WATCH OUT Implied volatility is extracted from today's option price. It is neither historical realized volatility nor a guaranteed forecast of future volatility.

Two variables: strike and maturity

For one underlying at a **fixed observation date** t, the market quotes a whole **implied volatility surface**:

$$\boxed{(K, T) \longmapsto \sigma_{\mathrm{imp},t}(K, T), \qquad T > t.}$$

Vary strike K, fix maturity T

One slice gives the **smile or skew**: how implied volatility changes across strikes for the same expiry.

Vary maturity T, fix strike K

Another slice gives a **term structure**: how implied volatility changes across expiries for the same strike.

Equivalently use time to maturity $\tau = T - t$. The subscript t records the observation date: the entire surface can change tomorrow.

Black–Scholes benchmark: a flat surface, with the same constant σ for every strike *and* maturity.

Reading the surface: a smile is only one slice

Illustrative annualized implied volatilities at today's spot $S_t = 100$:

Time to maturity	$K = 90$	$K = 100$	$K = 110$
3 months	28%	22%	20%
1 year	25%	21%	20%
2 years	24%	22%	21%

- Read a **row**: at 3 months, lower strikes have higher implied volatility (a downside skew).
- Read a **column**: at $K = 100$, implied volatility is $22\%, 21\%, 22\%$. The term structure need not be monotone.

Specify both coordinates

"The implied volatility is 22%" is incomplete without a strike and a maturity. A quoted 22% is annualized even for a 3-month option.

A downside skew is expensive crash insurance

For an equity index, low-strike puts are OTM yet often display higher implied volatility than ATM options.

Market observation

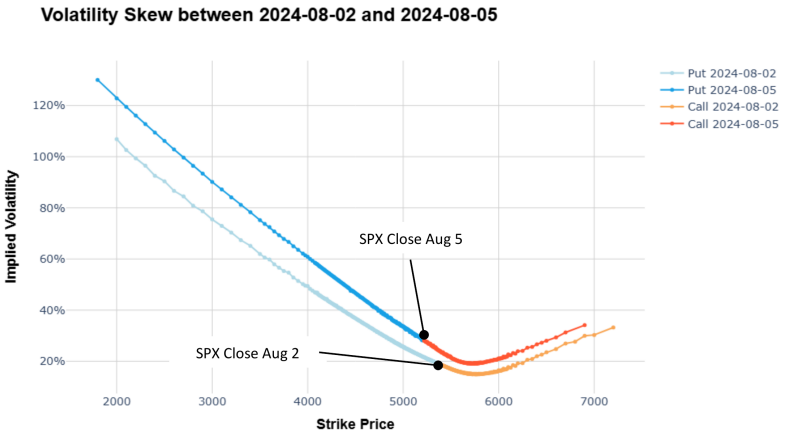
Investors demand protection against severe market falls, raising low-strike put prices.

Model reading

Feeding those high prices into Black–Scholes yields a high σ_{imp}: the volatility skew.

NO-ARBITRAGE The skew is not itself an arbitrage. It is evidence that constant-volatility Black–Scholes is an imperfect market description.

Market evidence: the S&P 500 downside skew



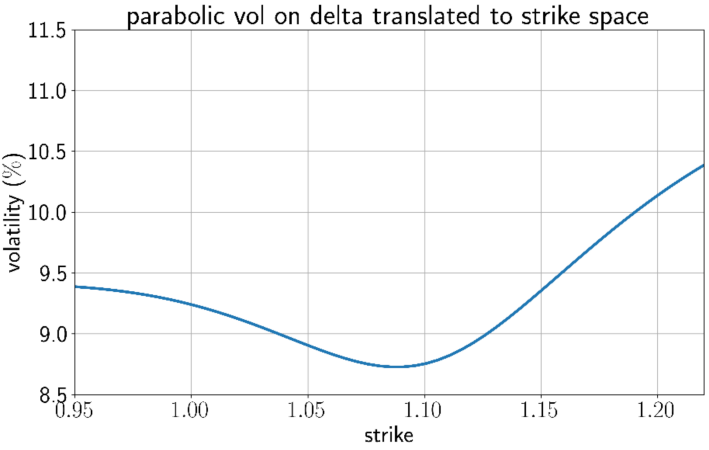
30-day tenor
2 and 5 August 2024.

Around the money, lower strikes have higher implied volatilities.

The curve also shifts between observation dates.

Source: Cboe, *The VIX Index Decomposition* (2025), Exhibit 7, p. 9. Market-based curves; 30-day tenor interpolated in variance between expiries.

Market evidence: a smile in EUR/USD options



6-month tenor
23 April 2025.

Both wings rise above the trough: a **smile**.

The smile need not be symmetric. Here the right wing is higher.

K: USD per EUR.

Source: U. Wystup, MathFinance (2025), Figure 2; ICE Data Services quotes. Author's parabolic interpolation in delta, converted to strike; reference spot fixed at 1.1000.

Model-Free Bounds

No-arbitrage already pins an option price inside an interval

Let $D_t = e^{-r\tau}$, still assuming no dividends.

Call

$$\boxed{\max(S_t - KD_t, 0) \leq C_t \leq S_t.}$$

Put

$$\boxed{\max(KD_t - S_t, 0) \leq P_t \leq KD_t.}$$

NO-ARBITRAGE These are model-free constraints. Black–Scholes prices must obey them, but so must every arbitrage-free model.

The call upper bound comes from a superhedge

One share always covers the call obligation:

$$S_T \geq (S_T - K)^+.$$

YOUR TURN Suppose $C_t > S_t$. What happens if you sell the call and buy the share? Track the cashflow today and the payoff at T.

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Arbitrage

Receive $C_t - S_t > 0$ today. At maturity, deliver the share if the call is exercised; otherwise retain a share worth $S_T \geq 0$. The terminal net payoff is nonnegative in every state.

Parity plus positivity gives the call lower bound

Put–call parity says

$$C_t = P_t + S_t - KD_t.$$

Since $P_t \geq 0$, $C_t \geq S_t - KD_t$. Since a call payoff is nonnegative, $C_t \geq 0$ as well. Together:

$$\boxed{C_t \geq \max(S_t - KD_t, 0).}$$

Same architecture as Lecture 8

Use a payoff inequality first; then translate it into a price inequality.

Are these quotes inside the arbitrage-free interval?

YOUR TURN Let $S_t = 100$, $K = 100$, $r = 5\%$, $\tau = 1$. Check call quotes 2, 10.45, and 103 against the model-free bounds.

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YOUR TURN Let $S_t = 100$, $K = 100$, $r = 5\%$, $\tau = 1$. Check call quotes 2, 10.45, and 103 against the model-free bounds.

$$\max(100 - 100e^{-0.05}, 0) \approx 4.88, \qquad C_t \leq 100.$$

Quote	2	10.45	103
Verdict	too low	feasible	too high

Superhedges and subhedges generalize the same logic

If a liquid strategy satisfies $Y_T \leq H \leq X_T$, then its initial costs give the interval

$$\boxed{Y_0 \leq \text{price}(H) \leq X_0.}$$

Claim too expensive

Sell H, buy the superhedge X, and keep the difference.

Claim too cheap

Buy H, short the subhedge Y, and keep the difference.

The Shape of an Arbitrage-Free Option Surface

A higher strike can never make the same call more valuable

For $K_1 < K_2$,

$$(S_T - K_1)^+ \geq (S_T - K_2)^+ \qquad \text{in every state.}$$

The dominance principle gives

$$\boxed{C_t(K_1) \geq C_t(K_2).}$$

For puts, the direction reverses

$$P_t(K_1) \leq P_t(K_2).$$

The right to sell at a higher price is more valuable.

Can you exploit a monotonicity violation?

YOUR TURN Two calls share the same maturity and have $K_1 = 90 < K_2 = 110$, but the market quotes $C_t(90) = 8$ and $C_t(110) = 10$. State the trade and explain why it cannot lose at maturity.

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Trade

Buy the lower-strike call for 8, sell the higher-strike call for 10, and receive 2 today. The terminal payoff

$$(S_T - 90)^+ - (S_T - 110)^+ \geq 0$$

in every state.

Convexity says call prices cannot bend the wrong way

For equally spaced strikes $K - h < K < K + h$, the butterfly payoff is

$$(S_T - K + h)^+ - 2(S_T - K)^+ + (S_T - K - h)^+ \geq 0.$$

Therefore its price must be nonnegative:

$$\boxed{C_t(K - h) - 2C_t(K) + C_t(K + h) \geq 0.}$$

NO-ARBITRAGE Equivalently, $K \mapsto C_t(K)$ is decreasing and convex. These are requirements on the entire strip of option quotes, not on one option alone.

Why is the butterfly payoff never negative?

YOUR TURN Let the strikes be 90, 100, and 110. Evaluate the butterfly payoff for $S_T \leq 90$, $90 < S_T < 100$, $100 \leq S_T < 110$, and $S_T \geq 110$.

Why is the butterfly payoff never negative?

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Region	payoff
$S_T \leq 90$	0
$90 < S_T < 100$	$S_T - 90$
$100 \leq S_T < 110$	$110 - S_T$
$S_T \geq 110$	0

It is a nonnegative tent, peaking at 10 when $S_T = 100$.

A three-quote diagnostic finds a butterfly arbitrage

Same maturity and underlying; the market quotes

$$C_t(90) = 14, \qquad C_t(100) = 12, \qquad C_t(110) = 7.$$

YOUR TURN Check monotonicity first, then convexity. If a condition fails, name the portfolio and compute its initial cashflow.

A three-quote diagnostic finds a butterfly arbitrage

Same maturity and underlying; the market quotes

$$C_t(90) = 14, \qquad C_t(100) = 12, \qquad C_t(110) = 7.$$

YOUR TURN Check monotonicity first, then convexity. If a condition fails, name the portfolio and compute its initial cashflow.

Monotonicity

$14 > 12 > 7$: no problem.

Convexity

Convexity needs $C_t(100) \leq \frac{1}{2}(14 + 7) = 10.5$, but $12 > 10.5$. The middle strike is overpriced relative to the wings.

Selling the expensive middle strike creates the arbitrage

Buy $C_t(90)$ and $C_t(110)$; sell two $C_t(100)$:

$$\text{initial cost} = 14 + 7 - 2(12) = \boxed{-3}.$$

The trader receives 3 today and holds the nonnegative butterfly payoff at maturity.

NO-ARBITRAGE

$$\text{payoff shape} \longrightarrow \text{price inequality} \longrightarrow \text{explicit trade.}$$

More time is valuable under a clean comparison

For a fixed strike, suppose $T_1 < T_2$, rates and dividends are zero. Then

$$\boxed{C_t(K, T_2) \geq C_t(K, T_1).}$$

Calendar-spread logic

If the longer call were cheaper, buy it and sell the shorter call. The trade generates cash today. At T_1, the remaining longer option is worth at least the intrinsic value needed to settle the short option.

WATCH OUT With rates or dividends, maturity comparisons require more care. Do not use this simple monotonicity statement without its assumptions.

Synthesis: Diagnose Before You Price

A disciplined order for checking an option quote

1. Identify the exact contract: underlying, strike, maturity, dividends.
2. Check positivity and the basic upper and lower bounds.
3. Pair calls and puts with the same K, T to check parity.
4. Across strikes, check monotonicity and convexity.
5. Across maturities, apply calendar checks only with the right assumptions.
6. If a check fails, write the portfolio before calling it an arbitrage.

NO-ARBITRAGE Black–Scholes provides a benchmark and an implied volatility; it never replaces model-free consistency checks.

Final challenge: build the trade, not just the inequality

Same maturity and stock; assume no dividends. The quotes are

$$S_t = 100, \quad K = 100, \quad r = 5\%, \quad \tau = 1, \quad C_t = 12, \quad P_t = 5.$$

YOUR TURN Is put–call parity satisfied? If not, write each leg of a zero-terminal-payoff arbitrage and its initial receipt.

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$$S_t = 100, \quad K = 100, \quad r = 5\%, \quad \tau = 1, \quad C_t = 12, \quad P_t = 5.$$

YOUR TURN Is put–call parity satisfied? If not, write each leg of a zero-terminal-payoff arbitrage and its initial receipt.

$$C_t + 100e^{-0.05} = 107.12 > 105 = P_t + S_t.$$

Sell the call, borrow 95.12, buy the put, and buy one share. The receipt is 2.12 today; parity makes the terminal net payoff zero.

The big picture in one line

option payoff geometry $\Longrightarrow$ arbitrage-free prices and hedges.

What to retain

Calls and puts are simple contingent claims. Replication gives parity; dominance gives bounds; strike convexity gives butterfly tests. Black–Scholes turns the same contracts into prices, Greeks, and implied volatilities.

NO-ARBITRAGE The habit to keep: every price statement should be backed by a payoff comparison or an explicit trading strategy.